

Notes: Chaderina, Weiss & Zechner (JFE, 2022)

The Maturity Premium

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Main CRSP-Compustat panel	3
2.2	Debt maturity measure	3
2.3	Firm characteristics used in portfolio interpretation	4
2.4	Portfolio construction	4
2.5	Financial crisis sample and Capital IQ debt maturity data	4
2.6	Leverage and beta dynamics	5
3	Models and empirical specifications	5
3.1	Cash-flow process	5
3.2	Stochastic discount factor and market price of risk	6
3.3	Debt and equity valuation	6
3.4	Debt issuance policy and the leverage ratchet effect	6
3.5	Equity beta and unconditional alpha	7
3.6	Portfolio and factor regression specification	7
3.7	Financial crisis difference-in-differences test	7
3.8	Leverage adjustment and conditional CAPM tests	7
4	Identification and theoretical strategy	8
4.1	Step 1: Establish a return premium from portfolio sorts	8
4.2	Step 2: Build a model that links maturity to state-dependent beta	8
4.3	Step 3: Use the 2008 crisis as a quasi-natural experiment	8
4.4	Step 4: Validate the dynamic mechanism in the full sample	8
4.5	Step 5: Test the conditional beta channel directly	9
4.6	Threats and limitations	9
5	Tables and figures	9
5.1	Table 1: Summary statistics and Table 2: Firm characteristics across debt maturity	9
5.2	Table 3: Debt maturity-sorted portfolios	10
5.3	Figure 1: Optimal rollover rate and Figure 2: Debt maturity and the leverage ratchet effect	11
5.4	Figure 3: Evolution of leverage and beta following cash-flow shocks	12
5.5	Figure 4: Beta, aggregate state, and market risk premium; Table 4: Maturity premium simulation	13
5.6	Figure 5: Maturity premium and parameters; Table 5: Benchmark simulation parameters	14

5.7	Table 6: Summary statistics for quasi-natural experiment and Table 7: Quarterly leverage difference-in-differences estimation	15
5.8	Figure 6: Quarterly leverage during the financial crisis and Figure 7: Beta during the financial crisis	16
5.9	Figure 8: Excess leverage and subsequent debt changes; Table 8: Leverage changes across excess leverage and debt maturity	17
5.10	Figure 9: Time dynamics of conditional betas and market price of risk; Table 9: Conditional CAPM	17
6	Interpretive synthesis	18
7	Short summary	18
8	Conclusion	19
8.1	Core conclusion	19
8.2	Economic intuition	19
8.3	Takeaway	19

1 Big picture

- **Question.** Does corporate debt maturity affect equity risk premia? In particular, do firms financed with long-term debt earn higher expected stock returns than otherwise similar firms financed with short-term debt?
- **Main empirical result.** A portfolio that is long firms with long debt maturities and short firms with short debt maturities earns a positive maturity premium of about 0.21% per month after controlling for unconditional market exposure.
- **Why maturity matters.** Debt maturity affects how quickly firms can adjust leverage after cash-flow shocks. Short-maturity firms can let maturing debt expire or roll over only part of it; long-maturity firms have more debt locked in place.
- **Mechanism.** Long-term debt creates sticky leverage. In downturns, shareholders are reluctant to repurchase outstanding debt because doing so transfers value to debtholders. Therefore, long-maturity firms remain highly levered exactly when the market price of risk is high.
- **Asset-pricing channel.** Long-maturity firms have betas that covary more positively with the market price of risk. An unconditional CAPM or standard factor model misses this conditional beta variation and therefore estimates a positive alpha.
- **Model contribution.** The paper embeds dynamic debt adjustment in an asset-pricing framework with a countercyclical market price of risk, generating a tractable link between maturity, leverage dynamics, beta dynamics, and expected equity returns.
- **Empirical support.** The paper combines portfolio sorts, simulations, a 2008 financial crisis quasi-natural experiment, leverage adjustment tests, and conditional CAPM estimates.
- **Economic takeaway.** Debt maturity is not just a corporate finance variable. It determines how leverage evolves over the business cycle, and that dynamic leverage exposure is priced in equity markets.

2 Data and measurement

2.1 Main CRSP-Compustat panel

- The main empirical panel matches CRSP stock returns with Compustat accounting data from January 1976 through December 2017.
- The unit of observation in the descriptive and portfolio analysis is the firm-month.
- Table 1 reports 1,840,640 firm-month observations for 18,392 unique firms.
- The main equity outcome is monthly excess return, measured relative to the risk-free rate.
- Firm size is market equity, measured in million USD. Book-to-market is the standard value proxy.
- Leverage is the ratio of debt to firm value, reported as a percentage.

Interpretation: The long sample makes the maturity premium a broad asset-pricing fact rather than a crisis-only phenomenon. The matched CRSP-Compustat design is also important because maturity comes from accounting disclosures, while returns come from stock-market data.

2.2 Debt maturity measure

- The main maturity variable is the fraction of debt that matures in more than three years.
- The paper follows the common corporate finance measure used by Custodio et al. (2013).

- Firms are sorted into five maturity buckets. The short-maturity bucket has very little debt maturing beyond three years; the long-maturity bucket has most debt maturing beyond three years.
- Table 2 shows that the shortest maturity bucket has about 5.18% of debt maturing after three years, while the longest maturity bucket has about 95.25%.

Interpretation: Debt maturity is measured as a balance-sheet maturity composition, not as a single bond maturity. This makes it suitable for a broad firm-level stock-return panel, but it also motivates later tests using Capital IQ debt issuance data, where remaining maturity around the crisis can be measured more directly.

2.3 Firm characteristics used in portfolio interpretation

- Market equity controls for the fact that smaller firms tend to use shorter debt maturities.
- Book-to-market checks whether the maturity premium is just another form of the value premium.
- Leverage controls for the level effect of debt on beta.
- Idiosyncratic volatility matters because short-maturity firms have higher default probabilities and higher rollover-risk exposure.
- Marginal tax rates affect target leverage in the model because tax benefits of debt make borrowing more attractive.
- Cash and CAPEX are included to describe liquidity and investment differences across maturity buckets.

Interpretation: Table 2 shows that maturity groups differ in size, leverage, volatility, and cash. This is why the paper uses size-maturity double sorts and model simulations calibrated to match characteristics of short- and long-maturity firms.

2.4 Portfolio construction

- Each month, firms are first sorted into five size quintiles.
- Within each size group, firms are then sorted into five debt-maturity portfolios.
- For each size bucket, the paper constructs a long-minus-short portfolio, long the highest maturity bucket and short the lowest maturity bucket.
- The overall LMS portfolio is the average of the five size-specific LMS portfolios.

Interpretation: The conditional double sort is central because maturity is correlated with firm size. Without conditioning on size, a maturity portfolio could accidentally capture a size premium rather than a maturity premium.

2.5 Financial crisis sample and Capital IQ debt maturity data

- For the quasi-natural experiment, the paper uses Capital IQ data on firms' debt instruments around the 2008 financial crisis.
- The sample focuses on firms that typically issue debt with maturities between five and seven years at origination.
- Firms are sorted by *remaining maturity* at the end of 2007, before the full crisis shock arrives.

- This design separates original maturity choice from the timing of debt rollover needs. Two firms may issue similar seven-year debt but face different remaining maturities at the onset of the crisis because they issued at different dates.

Interpretation: This is the paper’s most credible design for isolating leverage-dynamic effects of maturity. Remaining maturity at the crisis is partly driven by lumpy historical issuance timing rather than contemporaneous expectations about the crisis.

2.6 Leverage and beta dynamics

- Crisis leverage is measured quarterly for short- and long-remaining-maturity firms.
- Crisis beta is estimated using rolling-window CAPM regressions with daily returns and 60-day moving averages.
- In the full-sample leverage adjustment tests, excess leverage is defined relative to the past four-quarter average, and also using the Frank-Shen target-leverage approach.
- Conditional betas for the LMS portfolio are estimated using the conditional CAPM methodology of Boguth et al. (2011).
- The market price of risk is estimated using macro variables, following the logic in Choi (2013).

Interpretation: The paper does not rely only on return alphas. It directly checks whether the model’s state-variable mechanism is visible in leverage and beta dynamics.

3 Models and empirical specifications

3.1 Cash-flow process

The firm-level operating cash flow before interest and taxes is

$$Y_{i,t} = X_t I_{i,t}, \quad (1)$$

where X_t is aggregate productivity and $I_{i,t}$ is firm-specific idiosyncratic productivity.

The aggregate productivity process is

$$dX_t = \mu(X_t, t)X_t dt + \sigma_X X_t dW_{X,t}^P, \quad (2)$$

with mean-reverting growth rate

$$\mu(X_t, t) = \mu_0 - k [\log(X_t) - (\mu_0 - \sigma_X^2/2) t]. \quad (3)$$

Firm-specific idiosyncratic shocks satisfy

$$dI_{i,t} = \sigma_i I_{i,t} dW_{I,i,t}^P. \quad (4)$$

Interpretation: The cash-flow model generates both aggregate business-cycle risk and firm-specific risk. Debt maturity affects how leverage responds to these shocks; the aggregate component also makes the market price of risk move over the cycle.

3.2 Stochastic discount factor and market price of risk

The stochastic discount factor is

$$\frac{d\Lambda_t}{\Lambda_t} = -r dt - \frac{\mu(X_t, t) + \delta - r}{\sigma_X} dW_{X,t}^P. \quad (5)$$

The market price of risk is

$$\lambda_t = \frac{\mu(X_t, t) + \delta - r}{\sigma_X}. \quad (6)$$

The risk premium for bearing systematic risk is

$$\eta_t = \sigma_X \lambda_t. \quad (7)$$

Interpretation: Because aggregate productivity growth is mean-reverting, the market price of risk is countercyclical. This is the key state variable: leverage matters more in downturns because downturns are states in which risk compensation is high.

3.3 Debt and equity valuation

A firm issues debt with face value $F_{i,t}$. The maturity rate is m_i , so average maturity is $1/m_i$. Residual cash flows distributed to equityholders are

$$\Pi_{i,t} dt = \{Y_{i,t}(1 - \tau) + \tau c F_{i,t} - (c + m_i) F_{i,t} + G_{i,t} v_{i,t}^D\} dt. \quad (8)$$

The market value of equity is

$$V_{i,t}^E(Y_{i,t}, F_{i,t}) \Lambda_0 = E_0^P \left[\int_0^{t_b} \Lambda_s \Pi_{i,s} ds \right]. \quad (9)$$

Interpretation: The equity value reflects operating cash flows, taxes, coupons, maturing debt repayment, and new issuance or repurchase. Shareholders choose debt issuance dynamically, but cannot commit to future leverage reductions.

3.4 Debt issuance policy and the leverage ratchet effect

The scaled debt issuance policy is

$$g_{i,t}(y_{i,t}) = m_i \left(\frac{y_{i,t}}{y_{m,i}} \right)^{\gamma_i}, \quad (10)$$

where $y_{m,i}$ is the scaled cash-flow level at which debt issuance exactly equals maturing debt.

Market leverage is

$$L_{i,t} = \frac{v_{i,t}^D}{v_{i,t}^D + v_{i,t}^E}. \quad (11)$$

Interpretation: If cash flows fall below the threshold, shareholders do not fully roll over maturing debt. Short-maturity debt creates frequent rollover opportunities and therefore faster deleveraging. Long-maturity debt creates sticky leverage because little debt matures at each point in time.

3.5 Equity beta and unconditional alpha

The conditional expected equity return can be written as

$$E_t^P[r_{i,t}^E dt] = r dt + \beta_{i,t} \eta_t dt. \quad (12)$$

The model implies a leverage-based beta relation of the form

$$\beta_{i,t} = \left(1 + \frac{v_{i,t}^D}{v_{i,t}^E}\right) \beta^{Assets}. \quad (13)$$

Unconditional alpha from a CAPM can be written in terms of the covariance between conditional beta and the market price of risk:

$$\alpha_i = \left[1 - \frac{\eta^2}{\sigma_M^2}\right] Cov(\beta_{i,t}, \eta_t) - \frac{\eta}{\sigma_M^2} Cov(\beta_{i,t}, (\eta_t - \eta)^2). \quad (14)$$

Interpretation: This is the core asset-pricing equation. If a firm's beta is high exactly when the market price of risk is high, an unconditional CAPM misses this state dependence and produces a positive alpha. Long-maturity debt raises this covariance because leverage remains elevated in bad states.

3.6 Portfolio and factor regression specification

For the LMS maturity portfolios, the paper estimates regressions of the form

$$R_t^{LMS} = \alpha + \beta^M MKT_t + \varepsilon_t, \quad (15)$$

for CAPM alphas, and extended models with Fama-French factors:

$$R_t^{LMS} = \alpha + \beta^M MKT_t + \beta^{SMB} SMB_t + \beta^{HML} HML_t + \beta^{RMW} RMW_t + \beta^{CMA} CMA_t + \varepsilon_t. \quad (16)$$

Interpretation: The empirical maturity premium is an alpha relative to unconditional factor models. The model then explains why an alpha appears: not because markets are irrational, but because unconditional factor regressions omit conditional beta variation.

3.7 Financial crisis difference-in-differences test

The crisis leverage test estimates a specification of the form

$$Leverage_{i,t} = \alpha_i + \delta_t + \theta(Post_t \times LT_i) + u_{i,t}, \quad (17)$$

where LT_i indicates firms with longer remaining debt maturity at the end of 2007.

Interpretation: The identifying variation comes from remaining maturity at the onset of the crisis among firms that issue similar original maturities. A positive coefficient on $Post \times LT$ means that long-remaining-maturity firms experience larger post-crisis leverage increases.

3.8 Leverage adjustment and conditional CAPM tests

The paper checks the mechanism using two additional empirical structures:

- changes in debt following excess leverage, separately for short- and long-maturity firms;
- a conditional CAPM in which LMS beta is allowed to vary with conditioning variables and is compared with the market price of risk.

Interpretation: These tests move beyond the cross-sectional premium. They show that long debt is stickier, and that the LMS portfolio’s conditional beta comoves with the market price of risk.

4 Identification and theoretical strategy

4.1 Step 1: Establish a return premium from portfolio sorts

The first step is descriptive asset pricing: sort firms on debt maturity while controlling for size, then test whether long-maturity firms outperform short-maturity firms. The maturity premium is not interpreted as causal by itself. It is an empirical regularity that the model must explain.

Interpretation: The double-sort design removes the most obvious confound, because firm size is strongly related to debt maturity. Factor regressions further check whether the maturity premium is simply the value premium, size premium, profitability premium, or investment premium.

4.2 Step 2: Build a model that links maturity to state-dependent beta

The model identifies a mechanism: debt maturity affects the speed of leverage adjustment, which affects the covariance between conditional beta and the market price of risk. The model is calibrated so that short- and long-maturity firms resemble the empirical maturity buckets.

Interpretation: The simulation is not an independent empirical causal test, but it shows that the mechanism is quantitatively plausible. Table 4’s 0.19% simulated premium is close to the 0.21% empirical premium.

4.3 Step 3: Use the 2008 crisis as a quasi-natural experiment

The financial crisis creates a large negative cash-flow and financial-market shock. The paper compares firms that typically issue similar maturity instruments but differ in remaining maturity at the end of 2007.

- Treated group: firms with longer remaining maturity at the onset of the crisis.
- Control group: firms with shorter remaining maturity at the onset of the crisis.
- Key logic: remaining maturity depends on prior issuance timing and lumpiness, not only on current target maturity choice.
- Prediction: short remaining maturity should produce a sharper initial spike but faster deleveraging; long remaining maturity should produce persistent high leverage and beta.

Interpretation: This design addresses the concern that firms choose long maturity because they are fundamentally different. By restricting to firms with similar maturity at issuance and using remaining maturity at a specific shock date, the paper gets closer to exogenous variation in rollover timing.

4.4 Step 4: Validate the dynamic mechanism in the full sample

The paper then asks whether short-maturity firms actually adjust debt faster outside the crisis window. Fig. 8 and Table 8 show that short-maturity firms reduce debt more strongly when they are above target leverage.

Interpretation: This supports the core mechanism: maturity affects the speed of leverage correction, not only crisis-period outcomes.

4.5 Step 5: Test the conditional beta channel directly

Finally, Fig. 9 and Table 9 connect the asset-pricing implication to data. If the maturity premium reflects omitted conditional risk, then the beta of the LMS portfolio should comove with the market price of risk, and a conditional CAPM should remove the alpha.

Interpretation: This is the most direct bridge between corporate finance and asset pricing in the paper. It shows that the maturity premium is not just a static characteristic return; it is tied to time-varying risk exposure.

4.6 Threats and limitations

- Debt maturity may be jointly determined with firm risk, investment opportunities, or financial constraints.
- The Compustat maturity measure is coarse and based on a threshold of debt maturing after three years.
- The crisis experiment reduces, but does not completely eliminate, concerns about selection into maturity structures.
- The model abstracts from many institutional details, including seniority, covenants, transaction costs, endogenous maturity changes, and debt-market liquidity premia.
- Conditional beta and market price of risk are estimated rather than observed, which introduces measurement error.

Interpretation: The paper’s strength is the coherence of multiple pieces of evidence. The portfolio return fact, the model, crisis dynamics, leverage adjustment patterns, and conditional beta results all point to the same mechanism.

5 Tables and figures

5.1 Table 1: Summary statistics and Table 2: Firm characteristics across debt maturity

Read:

- Table 1 shows the main distribution of returns, size, book-to-market, leverage, and debt maturity. Mean debt maturity is 53.15%, with a median of 58.87%.
- Mean monthly excess returns are 0.92%, but the standard deviation is high at 15.64%, confirming substantial return volatility.
- Table 2 shows that short-maturity firms are smaller, more levered, more idiosyncratically volatile, and hold more cash.
- Short-maturity firms have 5.18% of debt maturing after three years; long-maturity firms have 95.25%.
- Book-to-market ratios are similar across maturity buckets, which helps separate maturity from value.

Interpretation: The descriptive statistics show that maturity is not random. Short-maturity firms differ from long-maturity firms in size, volatility, cash, and leverage. This is why the paper needs conditional size sorts, factor controls, and a model that calibrates short- and long-maturity firms to observed characteristics.

Economic message: Debt maturity is an economically meaningful firm characteristic. It is not merely a technical liability classification; it is associated with leverage flexibility, cash holdings, and idiosyncratic risk.

Table 1

Summary statistics. We compute the mean and the standard deviation, as well as the 25%, 50%, and 75% quantiles, on a monthly frequency for the cross-section of firm characteristics. The table presents time-series averages of the monthly statistics. Excess returns, leverage, and debt maturity are displayed in %, and market equity is in million USD. The underlying data set comprises matched observations from CRSP and Compustat from January 1976 until December 2017. In total, the panel consists of 1,840,640 firm-month observations of 18,392 unique firms.

	Mean	SD	Q ₂₅	Median	Q ₇₅
Excess Returns	0.92	15.64	-6.08	0.01	6.45
Market Equity (ME)	2,365.89	9,903.90	59.10	260.15	1,111.80
Book-to-Market Ratio (BM)	0.93	0.94	0.43	0.74	1.16
Leverage (L)	31.33	23.92	10.71	27.08	48.55
Debt Maturity (DM)	53.15	33.87	21.55	58.87	83.32

(a) Table 1: Summary statistics

Firm characteristics across debt maturity. We compute average characteristics across five debt maturity buckets. We sort firms into quintiles for every year in our sample. The table presents time-series averages of the annual characteristics within each bucket. Idiosyncratic volatility is estimated from rolling CAPM regressions on monthly returns with a five-year window. Market equity is in million USD. The marginal tax rates are based on Blouin et al. (2010). The underlying data set comprises matched observations from CRSP and Compustat over the time horizon January 1976 until December 2017.

Debt Maturity	Short	...	Medium	...	Long
Debt Maturity (%)	5.18	32.86	60.23	79.59	95.25
Market Equity	520.59	2,815.75	3,368.51	2,529.03	1,492.49
Book-to-Market	0.92	0.98	0.96	0.96	0.93
Leverage (%)	23.89	32.97	34.23	35.11	32.08
IVOL (%)	14.93	12.29	10.93	10.40	11.25
Tax Rate (%)	19.71	23.50	26.10	27.68	27.67
Cash	18.03	11.19	9.53	9.35	12.03
β^H	0.84	0.70	0.68	0.68	0.76
CAPEX	0.19	0.16	0.14	0.13	0.16

(b) Table 2: Firm characteristics across debt maturity

5.2 Table 3: Debt maturity-sorted portfolios

Read:

- Panel A shows average monthly excess returns for portfolios double-sorted on size and debt maturity.
- Within the smallest size group, long-maturity firms earn 0.92% and short-maturity firms earn 0.59%, producing an LMS return of 0.33%.
- Panel B shows CAPM alpha, Fama-French 3-factor alpha, and Fama-French 5-factor alpha for the LMS portfolios.
- The overall LMS CAPM alpha is 0.21% per month with a Newey-West t-statistic of 2.93.
- The FF3 alpha remains 0.15% and statistically significant; the FF5 alpha is 0.10% and marginally significant.

Interpretation: Table 3 is the paper's central empirical return result. The maturity premium is not just a raw return spread; it survives market beta controls and standard factor controls. The significant HML loading shows overlap with value, but the alpha remaining after factor controls shows that maturity is not subsumed by value.

Economic message: Firms financed with long-term debt earn a premium relative to firms financed with short-term debt. The rest of the paper explains why this premium can arise from dynamic leverage risk rather than mispricing.

Table 3
Debt maturity-sorted portfolios. Panel A shows the average excess return of the value-weighted portfolios. The last row represents the long-maturity minus short-maturity portfolios (LMS). The portfolios are formed by double sorts on size (five buckets at the 20%, 40%, 60%, and 80% percentiles) and debt maturity (five buckets at the 20%, 40%, 60%, and 80% percentiles) conditional within each size group. Panel B displays results for the LMS portfolios within each size bucket (LMS1 for small to LMS5 for large firms) and the average return of these five portfolios in the last column (LMS itself). The long-short portfolios are characterized by their alpha estimates from CAPM-regressions (α^{CAPM}), the three-factor model by Fama and French (1993) (α^{FF3}), and the five-factor model by Fama and French (2015) (α^{FF5}). Moreover, factor loadings for FF5 are shown. We report t-statistics based on standard errors following Neter and West (1987, 1994) in parentheses. The underlying data set comprises matched observations from CRSP and Compustat from January 1976 until December 2017.

Panel A: Portfolio Sorts					
		Size			
		Small	Medium	Large	
Debt Maturity	Short	0.59	0.71	0.72	0.81
	Medium	0.66	0.82	0.86	0.89
	Long	0.75	0.82	0.92	0.89
	LMS	0.86	0.89	0.99	0.93
Panel B: Long-Short Portfolios (long-minus-short debt maturity, LMS)					
	LMS1	LMS2	LMS3	LMS4	LMS5
α^{CAPM}	0.44*** (2.98)	0.32** (2.03)	0.22* (1.82)	0.04 (0.41)	0.03 (0.26)
α^{FF3}	0.35*** (2.69)	0.20 (1.44)	0.09 (0.82)	0.00 (0.04)	0.11 (0.94)
α^{FF5}	0.30** (2.27)	0.10 (0.59)	-0.06 (-0.60)	-0.11 (-1.22)	0.25** (2.57)
β^M	-0.08** (-1.99)	-0.04 (-1.17)	0.02 (0.75)	0.08*** (3.62)	-0.13*** (-4.45)
β^{RM}	-0.17*** (-3.09)	-0.23*** (-3.53)	-0.12*** (-2.76)	-0.01 (-0.28)	-0.09*** (-2.97)
β^{SMB}	0.36*** (5.44)	0.33*** (3.80)	0.33*** (5.25)	0.00 (0.08)	-0.13 (-1.32)
β^{HML}	0.20*** (2.73)	0.22 (1.51)	0.36*** (5.18)	0.18** (2.50)	-0.36*** (-3.55)
β^{MOM}	-0.21* (-1.89)	0.05 (0.46)	0.01 (0.07)	0.20** (2.21)	-0.16 (-1.41)

***p < .01, **p < .05, *p < .1

Table 3: Debt maturity-sorted portfolios

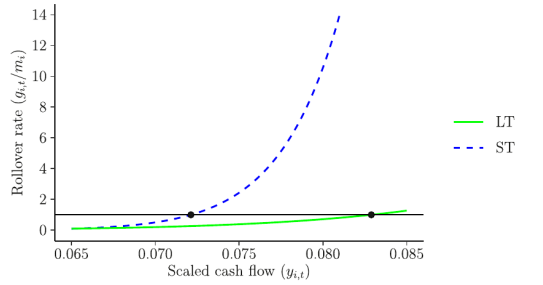
5.3 Figure 1: Optimal rollover rate and Figure 2: Debt maturity and the leverage ratchet effect

Read:

- Figure 1 shows that short-maturity firms have a much steeper rollover-rate function than long-maturity firms.
- At low scaled cash-flow levels, both firms reduce leverage, but short-maturity firms can do so faster because more debt matures.
- Figure 2 shows the same intuition dynamically. Short-maturity debt tracks cash flows more closely, while long-maturity debt adjusts slowly.
- Leverage in long-maturity firms rises more persistently after negative cash-flow realizations.

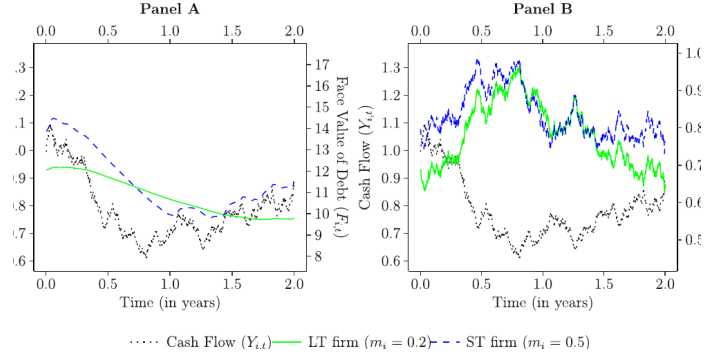
Interpretation: These figures are the core corporate finance mechanism. Long-term debt is sticky not because firms cannot issue or retire debt in principle, but because shareholders do not want to repurchase debt when cash flows are low. Short maturity provides an automatic adjustment channel through maturing debt.

Economic message: The maturity premium starts with a liability-side agency conflict. Shareholders' reluctance to reduce outstanding debt makes long-term leverage countercyclical.



er rate. This graph shows the optimal rollover rate of debt, which is given by the issuance policy $g_{i,t}$ scaled by the scaled cash-flow shock $y_{i,t}$. This ratio equals one when the firm's net issuance is zero. The short- and long-matur

(a) Figure 1: Optimal rollover rate



(b) Figure 2: Debt maturity and leverage ratchet effect

5.4 Figure 3: Evolution of leverage and beta following cash-flow shocks

Read:

- The figure compares short- and long-maturity firms after positive and negative cash-flow shocks.
- Short-maturity firms have a sharper immediate beta spike after a negative shock, but their leverage and beta decline quickly.
- Long-maturity firms have a more persistent elevation of leverage and beta after negative shocks.
- Positive shocks produce the mirror image: leverage falls, but adjustment again depends on debt maturity.

Interpretation: The figure clarifies an important nuance: short-maturity debt can look riskier at very short horizons because it creates rollover/default pressure after a sudden shock. The maturity premium studied in the paper is about longer-horizon persistence of risk exposure, where long debt keeps beta elevated in bad states.

Economic message: Maturity changes the timing of risk. Short debt concentrates risk around rollover shocks; long debt creates persistent state-dependent beta exposure.

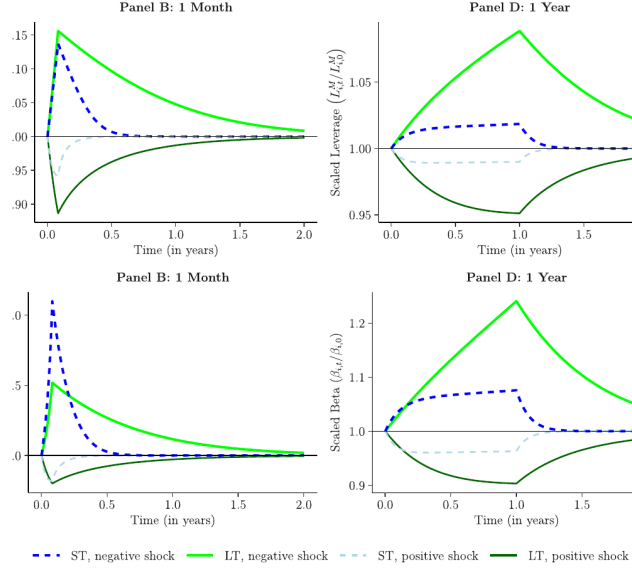


Figure 3: Evolution of leverage and beta following cash-flow shocks

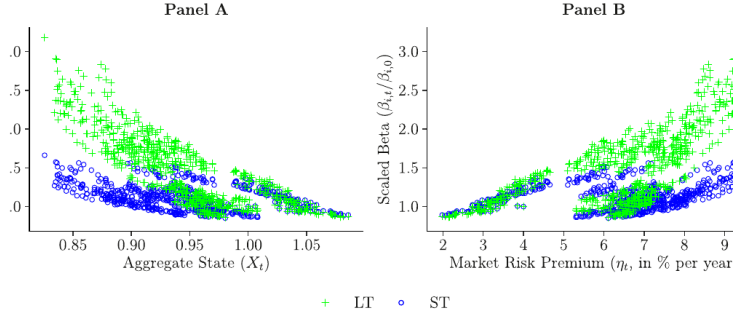
5.5 Figure 4: Beta, aggregate state, and market risk premium; Table 4: Maturity premium simulation

Read:

- Figure 4 shows scaled beta against the aggregate state and against the market risk premium.
- Long-maturity firms have higher beta when aggregate productivity is low and when the market risk premium is high.
- Table 4 calibrates two model firms to match short- and long-maturity characteristics from Table 2.
- The short-maturity simulated alpha is 0.06%, the long-maturity simulated alpha is 0.25%, and the simulated LMS maturity premium is 0.19% per month.

Interpretation: Figure 4 shows the state dependence behind the premium. Table 4 then demonstrates that the model can produce a maturity premium close to the empirical 0.21% estimate. The magnitude is not merely qualitative; it is quantitatively aligned with the data.

Economic message: Long debt is priced because it loads on bad states. It raises beta when risk compensation is high, not simply because it raises average leverage.



(a) Figure 4: Beta, aggregate state, and market risk premium. This figure shows the endogenous development of scaled beta ($\beta_{it}/\beta_{i,0}$) against the aggregate state in Panel A and against the market risk premium η_t in Panel B. The crosses (circles) represent $\beta_{it}/\beta_{i,0}$ for a firm with $\sigma_i = 0.10$ 2 ($m_i = 0.5$), i.e., a debt maturity of five (LT) and two (ST) years, respectively. The underlying parameters are as in the benchmark specification.

(a) Figure 4: Beta, aggregate state, and market risk premium

Table 4

Maturity premium. This table presents the simulation results for a panel of short-maturity and long-maturity financed firms that replicate the maturity premium observed in the data. The main parameters for this simulation are as in the benchmark specification, i.e., $\mu_0 = 0.05$, $k = 0.25$, $\sigma_X = 0.15$, $\delta = 0.04$, and $r = 0.05$. The parameters for the short- and long-maturity financed firms are chosen to reflect the characteristics of the firms in the lowest and highest maturity buckets; see Table 2.

	Short	Long	LMS
Debt Maturity (years)	1	5	
σ_i	20%	10%	
τ	20%	30%	
CAPM- α	0.06%	0.25%	
Maturity Premium			0.19%

(b) Table 4: Maturity premium simulation

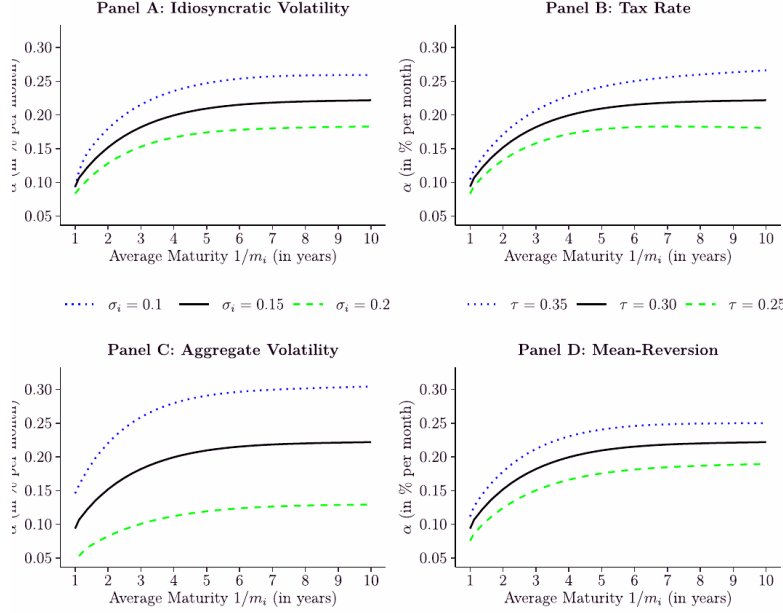
5.6 Figure 5: Maturity premium and parameters; Table 5: Benchmark simulation parameters

Read:

- Figure 5 shows simulated CAPM alphas as debt maturity varies from one to ten years.
- In every panel, alphas rise with maturity, especially from one to six years.
- Higher marginal tax rates raise the maturity premium because they support higher leverage.
- Higher aggregate volatility raises the premium because it increases variation in the market price of risk.
- Higher idiosyncratic volatility lowers the premium because firms choose lower leverage to avoid default risk.
- Table 5 lists benchmark parameters, including $\mu_0 = 0.05$, $k = 0.25$, $\sigma_X = 0.15$, $\sigma_i = 0.15$, $\tau = 0.30$, and maturities from one to ten years.

Interpretation: The comparative statics show that the maturity premium is strongest when leverage is high and aggregate risk prices vary strongly. The premium weakens when idiosyncratic risk is high because firms endogenously reduce leverage, muting the beta channel.

Economic message: Debt maturity is priced through an interaction between leverage choice and macro risk prices. Maturity matters more when firms have reasons to borrow and when recessions have high market prices of risk.



(a) Figure 5: Maturity premium and parameters

the simulation study, we group them into three categories. First, we present cash flow parameters associated with Y_{it} under both probability measures. Second, we show parameters used for three rates and debt-related parameters. The bottom line presents details on the simulation setting.

Cash Flow	μ_0 0.05	k 0.25	σ_k 0.15	σ_l 0.15	μ_Y 0.01
Rates & Debt	r 0.05	δ 0.04	τ 0.30	$1/m$ [1, 10]	c 0.07
Simulation	economies 10,000	firms 2500	years 10	Δt 1/1200	

(b) Table 5: Benchmark simulation parameters

5.7 Table 6: Summary statistics for quasi-natural experiment and Table 7: Quarterly leverage difference-in-differences estimation

Read:

- Table 6 describes the crisis subsample of firms with five- to seven-year maturities at issuance.
- Short and long remaining maturity firms have similar market equity, book-to-market, leverage, and idiosyncratic volatility in 2007.
- Table 7 estimates post-crisis leverage changes. The coefficient on $Post \times LT$ is 0.03 and statistically significant.
- The effect remains when firm and time fixed effects are included.

Interpretation: Table 6 supports comparability: the groups are not obviously different before the crisis except for debt maturity. Table 7 shows that firms with long remaining maturity experienced a larger leverage increase during the crisis, as predicted by sticky long-term debt.

Economic message: When the crisis hit, long remaining maturity worked like a slow deleveraging constraint. Firms with short remaining maturity could reduce debt more quickly through rollover decisions.

Table 6

Summary statistics for firms in quasi-natural experiment. This table summarizes characteristics of firms in the quasi-natural experiment as reported for the fiscal year 2007. We select firms with average maturities at issuance between five and seven years for instruments outstanding in December 2007. We sort firms into short (ST) and long (LT) maturity buckets based on the remaining debt maturity of the fiscal year 2007 at the 40% and 60% percentiles, respectively.

Debt Maturity	Short	Medium	Long
Number of Firms	108	54	108
Debt Maturity (%)	45.43	84.64	97.28
Market Equity	4,906.24	4,910.58	3,279.48
Book-to-Market	0.62	0.87	0.72
Leverage (%)	26.69	38.02	32.04
IVOL (%)	8.99	9.51	9.59

(a) Table 6: Summary statistics for quasi-natural experiment

Quarterly leverage difference-in-differences estimation. We regress quarterly leverage from the pre-period (the year 2007) and the post-period (2008 Q3 and Q4 as well as 2009 Q1 and Q2) on an interaction term for long-maturity financed firms at the onset of the financial crisis. In both models, firm fixed-effects are included. Model (2) adds fixed effects in the time dimension, which requires excluding the post-crisis dummy, which is included in Model (1).

	Model (1)	Model (2)
Post $\times$ LT	0.03*** (0.01)	0.03*** (0.01)
Post	0.14*** (0.01)	
FE	Firm	Firm, Time
Observations	1,490	1,490
Adjusted R ²	0.851	0.872

*** $n < .01$. ** $n < .05$. * $n < .1$

(b) Table 7: Quarterly leverage difference-in-differences

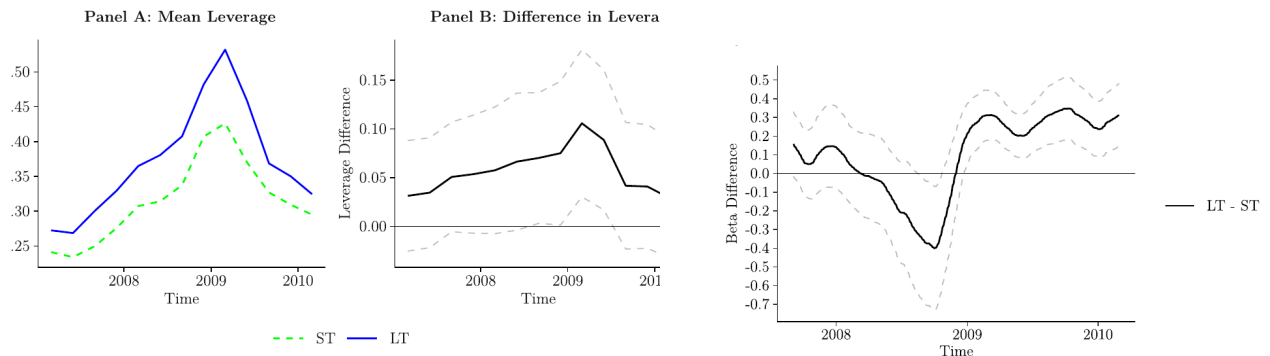
5.8 Figure 6: Quarterly leverage during the financial crisis and Figure 7: Beta during the financial crisis

Read:

- Figure 6 shows that both short- and long-maturity firms increased leverage during the crisis, but long-maturity firms increased more and remained elevated longer.
- The difference in leverage peaks around early 2009 and is statistically meaningful.
- Figure 7 shows the beta difference between long- and short-maturity firms.
- Short-maturity firms initially have higher beta, but the long-short beta becomes positive in 2009 and remains elevated as the market price of risk is high.

Interpretation: The crisis evidence matches the timing nuance in Figure 3. Short debt is risky immediately after a sudden shock, but long debt produces prolonged systematic exposure. This is the state dependence that generates the maturity premium over longer horizons.

Economic message: The financial crisis offers a real-world illustration of the model's mechanism: remaining maturity determines how quickly leverage and beta can move after a major aggregate shock.



Quarterly leverage during the financial crisis. This graph shows the development of quarterly leverage around the financial crisis. We sort firms into short (ST) and long (LT) maturity buckets based on the remaining debt maturity in September 2008 at the 40% and 60% percentiles, respectively. Panel A illustrates

(a) Figure 6: Quarterly leverage during the financial crisis

the financial crisis. We show the evolution of betas estimated in rolling windows over the past 60 days on daily returns at issuance between five and seven years for instruments outstanding in December 2007. We sort firms into

(b) Figure 7: Beta during the financial crisis

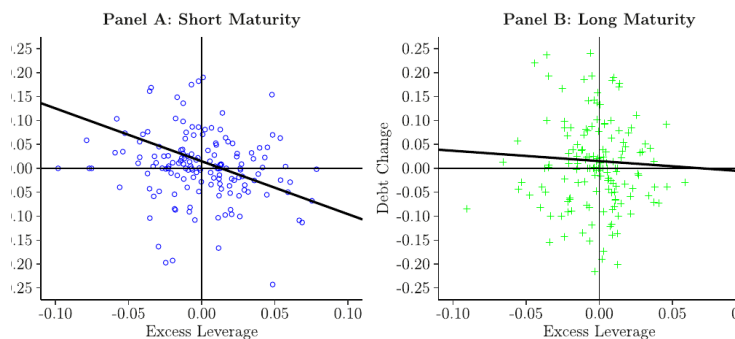
5.9 Figure 8: Excess leverage and subsequent debt changes; Table 8: Leverage changes across excess leverage and debt maturity

Read:

- Figure 8 compares debt changes following excess leverage for short- and long-maturity firms.
- Short-maturity firms show a steep negative relation: high excess leverage leads to debt reduction.
- Long-maturity firms show a much flatter relation, consistent with sticky debt.
- Table 8 uses the Frank-Shen excess leverage measure and finds that short-maturity firms with high excess leverage reduce leverage by -14.88, while long-maturity firms with high excess leverage reduce leverage by only -7.24.
- When leverage is low, short-maturity firms lever up more aggressively than long-maturity firms.

Interpretation: These tests validate the adjustment mechanism across the full sample, not only during 2008. Short maturity makes debt policy more responsive to deviations from target leverage. Long maturity dampens adjustment.

Economic message: Debt maturity affects the speed of capital structure rebalancing. This dynamic response is what turns maturity into an asset-pricing state variable.



ss leverage and subsequent debt changes. This figure shows excess leverage and subsequent relative quarterly changes in the face val
[Panel B], we show the results for firms within the shortest (longest) debt maturity quintile. We define excess leverage as the di
cane to each portfolio's average leverage over the next four quarters. Debt changes are relative to the next quarter. The solid line con

(a) Figure 8: Excess leverage and subsequent debt changes

leverage changes across excess leverage and debt maturity. This table shows relative leverage changes on an annual frequency in percent. We follow Frank and Shen (2019) and define excess leverage from firm-level regressions using the logarithm of firm assets and book-to-market ratios as predictors. We double-sort firms independently every year into buckets at the 33% and 66% breakpoints based on their excess leverage and debt maturity. Within the resulting groups, we take averages of the relative changes in leverage over the next year. The sample period is from January 1976 until December 2017.

		Excess Leverage		
		High	Medium	Low
Debt Maturity	Short	-14.88	13.44	46.65
	Medium	-7.74	6.59	28.78
	Long	-7.24	7.73	35.44

(b) Table 8: Leverage changes across excess leverage and debt maturity

5.10 Figure 9: Time dynamics of conditional betas and market price of risk; Table 9: Conditional CAPM

Read:

- Figure 9 plots the conditional beta of the LMS portfolio and an estimated market price of risk.
- The two time series move together, and the paper reports a statistically significant positive quarterly correlation of 0.48.
- Table 9 estimates a conditional CAPM for the maturity portfolios.
- The unconditional CAPM alpha for the LMS portfolio remains 0.21%, but the conditional alpha is 0.11 and statistically insignificant.
- The conditional beta terms are consistent with time-varying exposure to risk prices.

Interpretation: This is the strongest asset-pricing evidence for the model. If the maturity premium is compensation for conditional beta exposure, then allowing beta to vary with state variables should reduce the unconditional alpha. Table 9 shows exactly that.

Economic message: The maturity premium is not a puzzle under a conditional CAPM. It is a premium for being exposed to market risk in bad times.

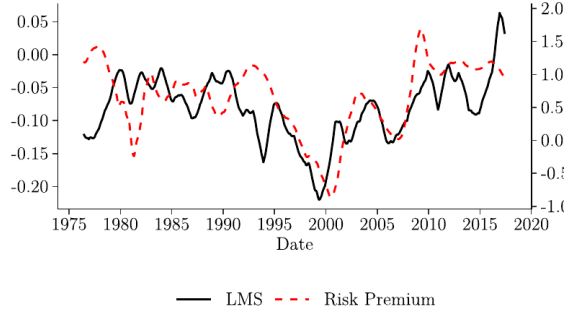


Figure 9: Conditional betas and the market price of risk. This figure shows the dynamics of conditional betas of the LMS following [Boguth et al. \(2011\)](#) (solid line on the left axis). We overlay the dynamics of the market price of risk (dashed line on the right axis).

(a) Figure 9: Conditional betas and market price of risk

Table 9
Debt maturity-sorted portfolios – conditional CAPM. The portfolios are formed by double sorts on size (five buckets at the 20%, 40%, 60%, and 80% percentiles) and debt maturity (five buckets at the 20%, 40%, 60%, and 80% percentiles) conditional within each size group. This table displays results for the five maturity portfolios by averaging across the size buckets (MAT1 for short-maturity to MAT5 for long-maturity financed firms) and the long-short portfolio of the difference between long-maturity and short-maturity financed firms (LMS). In the first row, we show CAPM intercept estimates for each portfolio. In the rows below, we show the alpha alongside other coefficients from the conditional CAPM following [Boguth et al. \(2011\)](#). We report t-statistics based on standard errors following [Newey and West \(1987, 1994\)](#) in parentheses. The underlying data set comprises matched observations from CRSP and Compustat from January 1976 until December 2017.

	MAT1	MAT2	MAT3	MAT4	MAT5	LMS
α^{CAPM}	-0.06 (-0.41)	0.09 (0.64)	0.16 (1.25)	0.21 (1.56)	0.15 (1.24)	0.21*** (2.93)
α^{COND}	0.04 (0.27)	0.12 (0.89)	0.16 (1.35)	0.20* (1.66)	0.15 (1.25)	0.11 (1.52)
β^M	1.29*** (4.22)	0.85*** (3.37)	0.92*** (4.00)	0.78*** (3.95)	1.14*** (4.54)	-0.04*** (-2.64)
Dividend Yield	0.03 (0.20)	0.17 (1.02)	0.32** (2.12)	0.28** (2.11)	0.12 (1.02)	0.07* (1.71)
Term Spread	-3.40 (-0.83)	-0.76 (-0.18)	-1.16 (-0.27)	-1.71 (-0.44)	-2.53 (-0.59)	-0.69 (-0.86)
T-Bill Rate	-1.73 (-0.67)	-0.50 (-0.21)	-1.69 (-0.76)	-2.06 (-0.96)	-1.20 (-0.56)	-0.81* (-1.90)
Default Spread	6.17 (0.87)	0.44 (0.05)	-1.78 (-0.23)	2.14 (0.34)	0.75 (0.11)	-3.79 (-1.49)
Beta LC36	-0.44 (-1.32)	-0.24 (-0.75)	-0.26 (-0.77)	-0.27 (-0.75)	-0.69* (-1.89)	-0.07 (-0.21)
Beta LC6	0.44 (1.00)	0.59* (1.76)	0.44 (1.30)	0.64 (1.60)	0.87** (2.23)	0.98** (2.44)
M36	-0.00* (-1.79)	-0.00 (-0.98)	-0.00 (-0.52)	-0.00 (-0.77)	-0.00* (-2.00)	-0.00 (-1.27)
M6	0.01*** (2.72)	0.01** (2.35)	0.01** (2.21)	0.01* (1.95)	0.01*** (2.74)	-0.00 (-0.46)

***p < .01; **p < .05; *p < .1

(b) Table 9: Conditional CAPM

6 Interpretive synthesis

- **The empirical premium.** The paper first establishes a cross-sectional maturity premium: long debt maturity is associated with higher risk-adjusted returns.
- **The corporate finance mechanism.** Long-term debt makes leverage sticky because shareholders do not want to repurchase debt after negative shocks.
- **The asset-pricing mechanism.** Sticky leverage raises beta in downturns, and downturns are precisely when the market price of risk is high.
- **The key distinction.** Short debt can create rollover risk at short horizons, but long debt creates persistent exposure to aggregate bad states.
- **Why unconditional models miss it.** Unconditional factor models control for average beta, not for covariance between beta and the market price of risk.
- **Why the evidence is coherent.** Portfolio returns, simulations, crisis leverage, beta dynamics, leverage adjustment tests, and the conditional CAPM all support the same maturity-risk channel.

7 Short summary

- The paper studies whether debt maturity affects equity risk premia.
- Firms with long debt maturities earn a positive maturity premium relative to firms with short debt maturities.
- The main empirical estimate is about 0.21% per month for the LMS maturity portfolio.
- The premium survives standard factor controls and is distinct from the value premium.
- The model shows that long-term debt creates sticky leverage through a leverage ratchet effect.

- In downturns, shareholders of long-maturity firms do not repurchase debt quickly, so leverage stays elevated.
- Elevated leverage raises equity beta exactly when the market price of risk is high.
- This covariance between beta and the market price of risk creates an unconditional CAPM alpha.
- Simulations calibrated to empirical short- and long-maturity firms generate a 0.19% monthly premium, close to the empirical 0.21%.
- The 2008 crisis experiment shows that long remaining maturity firms experienced more persistent leverage and beta increases.
- Full-sample leverage tests show that short-maturity firms adjust debt faster following excess leverage.
- Conditional CAPM estimates show that the maturity premium is reduced once beta is allowed to vary with the market price of risk.

8 Conclusion

8.1 Core conclusion

Chaderina, Weiss, and Zechner show that corporate debt maturity has asset-pricing consequences. Long-maturity debt makes leverage more countercyclical, which makes equity beta comove more with the market price of risk. This state-dependent beta exposure generates a maturity premium in unconditional asset-pricing tests.

8.2 Economic intuition

The intuition is that shareholders cannot commit to future leverage reductions. When cash flows fall, repurchasing debt benefits remaining debtholders and hurts equityholders, so long-maturity firms do not delever quickly. Short-maturity firms have a natural deleveraging channel because debt matures and need not be fully rolled over. Therefore, long-maturity firms remain more levered during bad times, and their equity becomes riskier in states where investors demand high compensation for risk.

8.3 Takeaway

The paper connects dynamic capital structure to cross-sectional asset pricing. Debt maturity determines not only rollover needs and default risk, but also the cyclicity of leverage and beta. A portfolio's alpha can therefore reflect omitted conditional risk rather than mispricing. The maturity premium is a priced consequence of slow leverage adjustment in bad states.